SIMATS ENGINEERING
ASSESSMENT TOOL 2:

Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Total Marks: 30

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Code of Conduct

I THIRU KUMARAN S / 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 4:

Smart Retail Inventory Management System

1) Cloud Architecture:

* POS systems and dashboards connect to centralized cloud services through APIs.
* Cloud services manage inventory and process store transactions.

2) Cloud Storage:

* Stores product details, transaction logs and analytics datasets.
* Backup and replication improve data safety and availability.

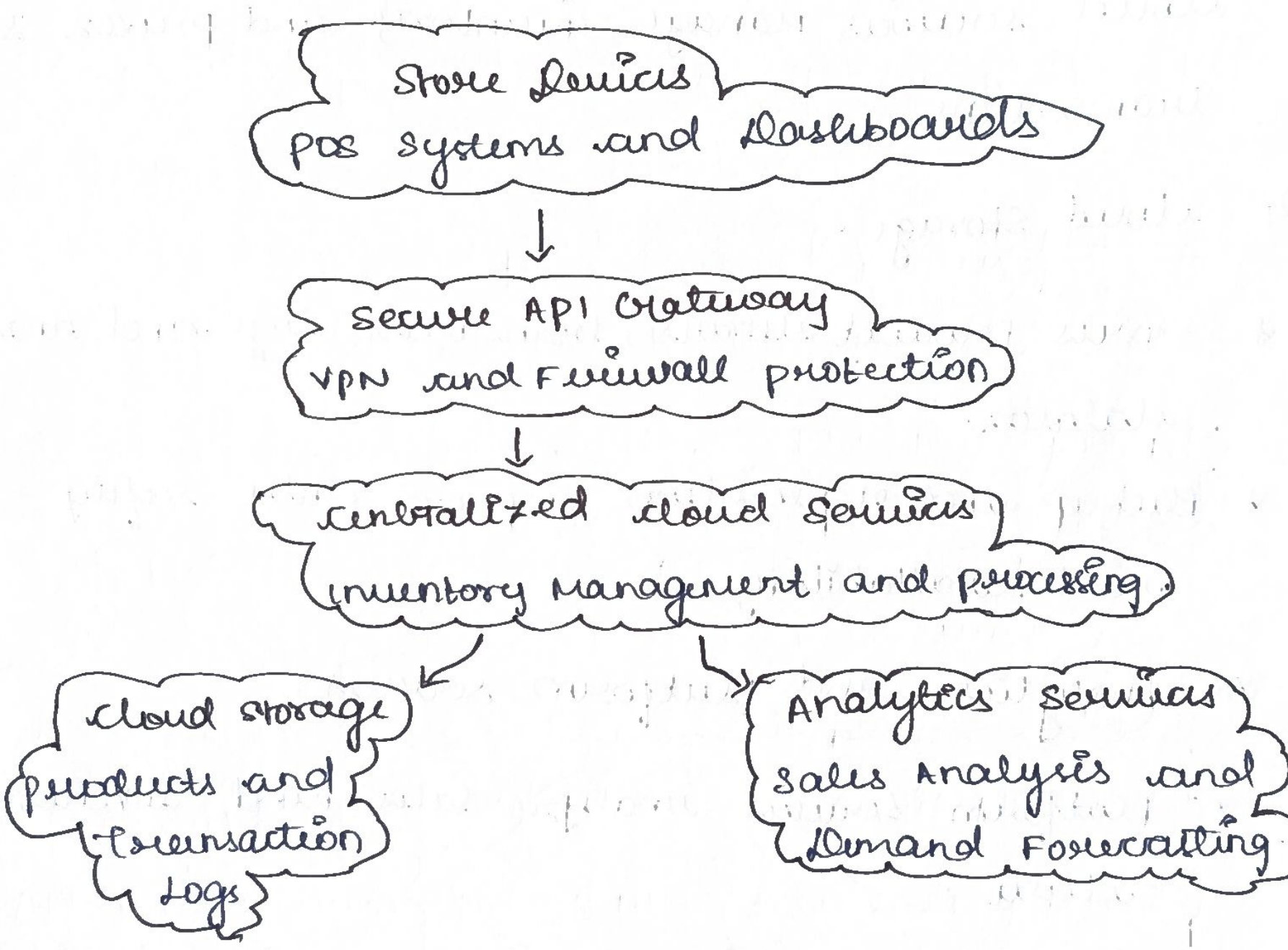
3) Analytics and platform services

* Platform services analyze sales and inventory data.
* Demand forecasting predicts product requirements and prevents stock shortages.

4) Security and scalability

* VPNs, firewalls, and APIs gatways provide secure access.
* Scalable cloud resources handle increasing transactions efficiently.

Architecture



⇒ overall, the system enables secure inventory management, accurate demand forecasting and efficient retail operations.

Case study 2: Cloud Based Document collaboration System

1) Cloud Architecture

* users access the collaboration platform through web browsers.
* cloud services support real time document editing and sharing.

2) Cloud storage

* Distributed cloud storage stores documents securely.
* version control maintains document history and supports recovery.

3) APIs and Network Infrastructure

* web APIs synchronize documents across the clients and devices.

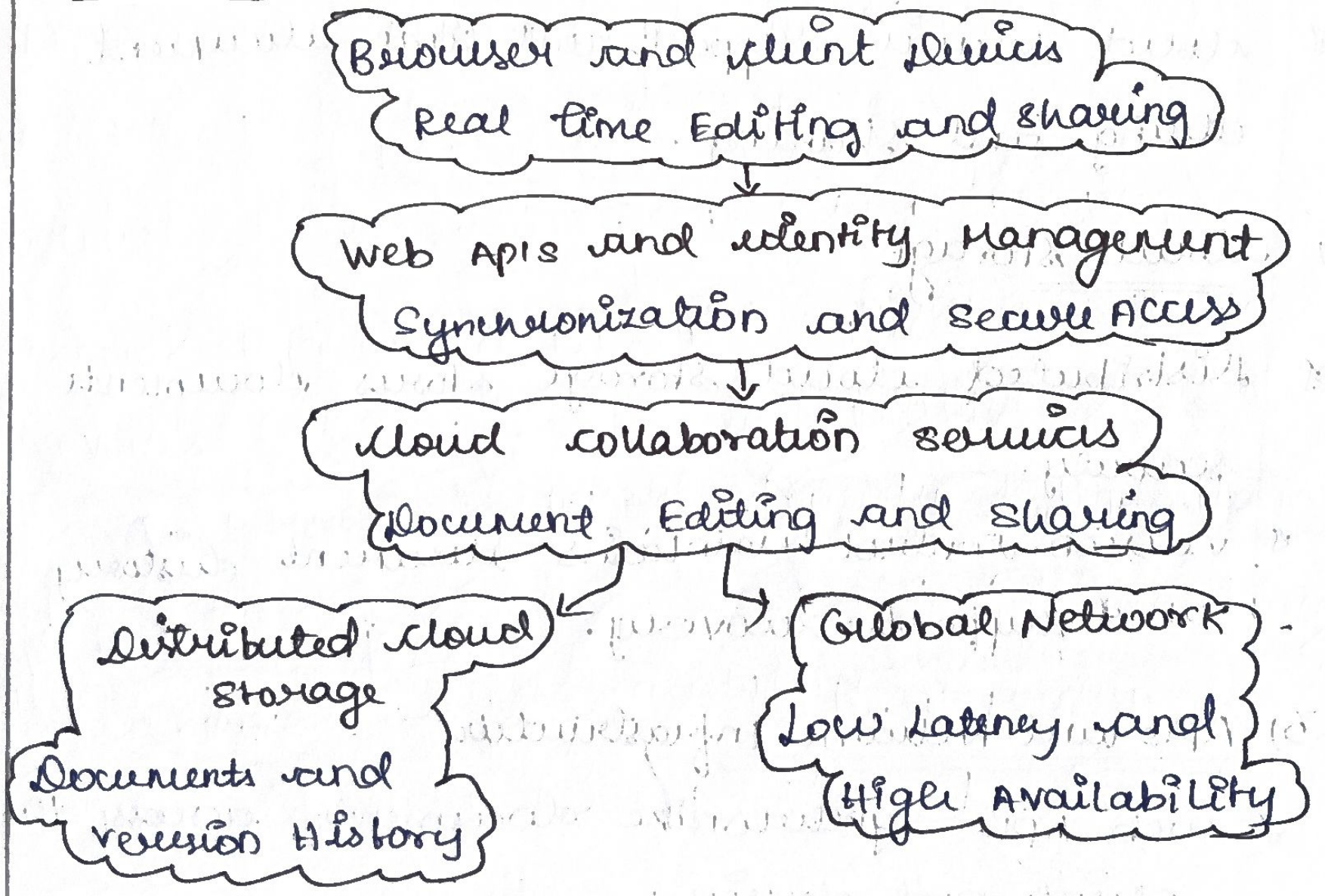
* Global Network Infrastructure ensures low latency and high availability.

* Security and Access control.

* Encryption protects documents during communication and storage.

* Identity management and access permissions secure collaboration.

Architecture:



∴ The system enables secure real time collaboration, reliable document storage and synchronization across multiple devices.

Case Study 5: Online Food ordering platform

(i) **Cloud Architecture:**

- web browsers and mobile clients access the food ordering platform.
- RESTful web APIs connect frontend clients with cloud backend services.

(ii) **Cloud Storage:**

- cloud storage securely stores menu images and invoices.
- customer data is stored securely for platform operations.

(iii) **Infrastructure and Scalability**

- Virtual Machines host backend services.

* Managed Kubernetes improves scalability and reliability.

4, Security

* Authentication tokens verify user identity.
* HTTPS and role based access control protect communication and data access.

Architecture

